

PPOL 6816 - Social Network Analysis

Syllabus – Subject to Change – Last Updated: 1/5/2025

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Course Description:

Social network analysis is primarily concerned with quantitatively summarizing the connections that exist between different social entities (usually people, but sometime organizations, nations, families, etc.), and measuring how individuals are shaped by those connections.

Network analysis can shed light on a variety of social problems and their trajectories, such as the origins of social segregation, the balance of geopolitical alliances, the structure of organizations, the spread of diseases and rumors, and the emergence of political polarization.

In this class, you will receive an introduction to different ways of analyzing social networks, and the questions, puzzles, and theories that are of interest to social network analysts. You will also begin to develop the core technical skills necessary for network analysis, by building network datasets, modeling network phenomena, and creating network visualizations.

Weekly Schedule:

Lectures: Thursday, 6:30-9pm; 125 E Street Room 360

Office Hours: Online, by appointment. On most class nights I will plan to stick around class afterwards to answer further questions.

Learning Objectives:

By the end of this course, you should:

1. Know basic network vocabulary and metrics.
2. Be able to collect and structure data that is suitable for network analysis.
3. Understand the mechanisms that guide the formation of social networks, and to be familiar with different ways of modeling and summarizing the structure of networks.
4. Have a sense for the social processes that social networks facilitate, and be familiar with different ways of modeling network influence and diffusion processes.
5. Feel comfortable critically evaluating research that draws on network analysis methods.
6. Know how the networks perspective can help us better understand broader social phenomena.

Prerequisites:

If you are unfamiliar with quantitative data analysis and programming, you should be prepared to work harder to pick these things up. Familiarity with basic concepts in probability, statistics, discrete math, and linear algebra will be helpful, but are by no means required.

Computational components of the course will use the R coding language. I will use RStudio as an environment for R and recommend that you do the same. (We also may do some exercises in NetLogo, a program used for creating and visualizing social simulations, but these will not require coding.)

Materials and Readings:

Textbooks:

The course textbook is *Analyzing Social Networks Using R*, by Stephen P. Borgatti, Martin G. Everett, Jeffrey C. Johnson, and Filip Agneessens. It is available on Amazon both digitally and in print.

Some readings will come from *Networks, Crowds, and Markets*, which is available in its entirety online: <https://www.cs.cornell.edu/home/kleinber/networks-book/>

Other Readings:

Additional readings will be posted to Canvas. Most required readings outside of the two books will be shorter or written for non-academic audiences (newspapers, magazines, blogs, etc.).

Academic Article Presentations:

In addition, 2-3 research articles will be posted for each week, which individual students will present to opening the following week's class. Each student will present 2 of these articles each throughout the semester. Completing this will be part of your participation grade.

You are not required to summarize the entire article you are presenting, but rather, share the parts of the article that you found most interesting or relevant to what we've discussed in class. Ideally, your presentation will last between 5-7 minutes and give the class material to build on for discussion. Students should prepare around 2-3 slides for their presentation (or an alternative presentation method/tool), and submit to Canvas before they present.

Other students are not required to read the entirety of these articles, but quickly skimming them may help you participate in class discussion.

Supplemental Articles:

Other articles will be posted on Canvas are suggested and meant to help guide you through the concepts and ideas we discuss in class, and give interested students the chance to explore certain topic areas in greater depth. They are not required.

Class Sessions:

These will be a combination of lectures, discussions, and in-class exercises. Please bring your computer to participate in class exercises – these will help you get more 'hands on' experience with network analysis, and will often mirror questions from problem sets.

Grading and Assessment:

The course grading will consist of problem sets, a single midterm examination, an independent network analysis project, and participation.

Problem Sets: 45%

There will be 5 assignments over the course of the semester. They will be assigned 2 weeks from the time they are due. Most will involve a combination of data analysis, coding, quantitative reasoning, and thinking critically about network problems and data. Canvas will be used for all submitted work.

Participation/Attendance: 20%

I am looking for evidence that you are engaging with the class – this means showing up and contributing on a regular basis. This part of the grade will also include class presentation of research articles.

Independent Network Analysis: 35%

You will build a small dataset on a topic of your own choosing, construct a network(s), perform analysis and visualization of your choice, and briefly present your findings to the class. I will ask for you to submit your topic and data earlier in the semester to make sure you are on the right track. During the final class session you will present your data and preliminary findings to the class.

Late Work

I understand that this class is not the only priority in your life and will try to be as flexible as possible. That said, deadlines are important in terms of keeping pace with the material, and in protecting both your time and mine. It is important that you are **extremely** proactive about communicating with me if you are having trouble with completing assignments on time. In the absence of an acceptable excuse or instructor permission, I will penalize work late work by 10% after the first day, 15% after three days, 20% after a week, and an additional 20% for each additional week thereafter.

Grade Distributions:

I may lower these thresholds as the semester goes on (I will not raise any number below), but at a minimum the following numerical averages will correspond to the following grades:

Grade	Range
A	95% or Higher
A-	91% - 94.99%
B+	87% - 90.99%
B	83% - 86.99%
B-	79% - 82.99%
C	70% - 78.99%
F	Less than 70%

Academic Resource Center/Disability Support: If you believe you have a disability, contact the Academic Resource Center (arc@georgetown.edu) for further information. The Center is located in the Leavey Center, Suite 335 (202-687-8354). The Academic Resource Center is the campus office responsible for reviewing documentation provided by students with disabilities and for determining reasonable accommodations in accordance with the Americans with Disabilities Act (ASA) and University policies. For more information, go to <http://academicsupport.georgetown.edu/disability/>.

Important Academic Policies and Academic Integrity: McCourt School students are expected to uphold the academic policies set forth by Georgetown University and the Graduate School of

Arts and Sciences. Students should therefore familiarize themselves with all the rules, regulations, and procedures relevant to their pursuit of a Graduate School degree. The policies are located at: <http://grad.georgetown.edu/academics/policies/>

Collaboration and Integrity

For this class specifically, while you are encouraged to consult with one another and help one another, you should each be completing the assignments on your own. This means that while you can discuss an assignment with your peers, you should not be sharing snippets of code or written answers to questions. If an assignment involves building a dataset, you should not be submitting the same data as someone else. **If you are consulting with someone, please list their names on your assignment.**

Use of Artificial Intelligence and Large Language Models:

If you are using AI/LLM tools such as ChatGPT, I expect that they are used like Wikipedia, Google, or any other technology that can aid with understanding and research. Cite and rephrase work that you did not create and verify that the solutions or sources that you have found are correct by conducting further research. Furthermore, you are responsible for any errors that these technologies might generate when assisting you. **If you are using AI tools in an assignment, presentation, or paper, please include 1-2 sentences to your submission describing how you've used them.**

Provost's Policy Accommodating Students' Religious Observances:

Georgetown University promotes respect for all religions. Any student who is unable to attend classes or to participate in any examination, presentation, or assignment on a given day because of the observance of a major religious holiday or related travel shall be excused and provided with the opportunity to make up, without unreasonable burden, any work that has been missed for this reason and shall not in any other way be penalized for the absence or rescheduled work. Students will remain responsible for all assigned work. Students should notify professors in writing at the beginning of the semester of religious observances that conflict with their classes. The Office of the Provost, in consultation with Campus Ministry and the Registrar, will publish, before classes begin for a given term, a list of major religious holidays likely to affect Georgetown students. The Provost and the Main Campus Executive Faculty encourage faculty to accommodate students whose bona fide religious observances in other ways impede normal participation in a course. Students who cannot be accommodated should discuss the matter with an advising dean.

Title IX/Sexual Misconduct

Georgetown University and its faculty are committed to supporting survivors and those impacted by sexual misconduct, which includes sexual assault, sexual harassment, relationship violence, and stalking. Georgetown requires faculty members, unless otherwise designated as confidential, to report all disclosures of sexual misconduct to the University Title IX Coordinator or a Deputy Title IX Coordinator. If you disclose an incident of sexual misconduct to a professor in or outside of the classroom (with the exception of disclosures in papers), that faculty member must

report the incident to the Title IX Coordinator, or Deputy Title IX Coordinator. The coordinator will, in turn, reach out to the student to provide support, resources, and the option to meet. [Please note that the student is not required to meet with the Title IX coordinator.]. More information about reporting options and resources can be found on the Sexual Misconduct Website: <https://sexualassault.georgetown.edu/resourcecenter>.

If you would prefer to speak to someone confidentially, Georgetown has a number of fully confidential professional resources that can provide support and assistance. These resources include:

Health Education Services for Sexual Assault Response and Prevention: confidential email sarp@georgetown.edu

Counseling and Psychiatric Services (CAPS): 202.687.6985 or after hours, call (833) 960-3006 to reach Fonemed, a telehealth service; individuals may ask for the on-call CAPS clinician

More information about reporting options and resources can be found on the [Sexual Misconduct Website](#).

Tentative Schedule and Topics (As of 1/5/2025)

1/9: Course Overview/Intro to Network Models and Measures

Readings:

Excerpt from *Connected* (Christakis and Fowler)

1/16: Models and Measures Continued, Centrality Measures

Problem Set 1 Assigned

Required Readings:

Your Network's Structure Matters More than Its Size (HBR – Libert)

Why Your Friends Have More Friends Than You Do (Nautilus – Holden)

Borgatti et. al Chapter 1 (Introduction)

Borgatti et. al Chapter 2 (Mathematical Foundations)

Borgatti et. al Chapter 10 (Centrality)

Supplemental Readings (Optional):

Borgatti 2005 - Centrality and Network Flow

Articles for Presentation (on 1/23):

Goncalves et. al 2017 – Exploring Team Passing Networks and Player Movement Dynamics in Youth Association Football

Ilany and Akcay 2016 – Social inheritance can explain the structure of animal social networks

1/23: Ego Networks and Social Capital

Required Readings:

The Loneliness Scare – Boston Review (Fischer)

The Decline of Trust and Neighborliness - Institute for Family Studies (O'Rourke)

Borgatti et. al Chapter 15 (Ego Networks)

Articles for Presentation (on 1/30):

Marsden 1987 - Core Discussion Networks of Americans

Lam et. al 2017 - Depression, Smoking, and Ego-Centric Social Network Characteristics in Ohio Appalachian Women

1/30: Mechanisms of Tie Formation and Homophily:

Problem Set 1 Due

Problem Set 2 Assigned

Required Readings:

Friends are more Genetically Similar than Strangers (Time – Ducharme)

Chapter 4 - Easley and Kleinberg (NCM)

Supplemental Readings (Optional):

McPherson et al 2001

Articles for Presentation (on 2/6):

Potarca 2017 - Does the internet affect assortative mating? Evidence from the US and Germany

Lozares et. al 2017- Homophily and heterophily in personal networks. From mutual acquaintance to relationship intensity

2/6: Triangles, Stars, and Exponential Random Graph Models

Required Readings:

Making Friends of Friends Benefits Hyenas

Borgatti et. al Chapter 15 (Only Parts on ERGMs)

Articles for Presentation (on 2/13):

Goodreau et 2009 (Focus on part through p. 108).

McMillan and Felmlee 2020 Beyond Dyads and Triads: A Comparison of Tetrads in Twenty Social Networks

2/13: Preferential Attachment and Core-Periphery Networks; Multiplex/Weighted/Signed Networks

Problem Set 2 Due

Problem Set 3 Assigned

Required Readings:

Your Work Friends and Enemies are Affecting Your Performance
(KelloggInsight – Meier)

Going viral? Study examines the probability of social media posts hitting it big –
(TechXplore – Randall 2022)

Chapter 5 - Easley and Kleinberg (NCM)
Chapter 18 – Easley and Kleinberg (NCM)

Suggested Readings (Optional):

Barabasi and Bonabeau 2003
Barabasi and Albert 1999

Articles for Presentation (on 2/20):

Kinne 2014 - Dependent Diplomacy: Signaling, Strategy, and Prestige in the Diplomatic Network

Maoz 2012 - Preferential Attachment, Homophily, and the Structure of International Networks, 1816-2003

2/20: Small-Worlds, Long Ties, and Random Walks

Required Readings

Proof! Just six degrees of separation between us (The Observer – Smith 2008)

A Massive LinkedIn Study Reveals Who Actually Helps You Get That Job
(Scientific American – Callier 2022)

Chapter 3 – Easley and Kleinberg (NCM)
Chapter 20 – Easley and Kleinberg (NCM)

Suggested Readings (Optional):

Travers and Milgram 1969
Watts and Strogatz 1998
Watts 1999 (Skim)

Articles for Presentation (on 2/27):

Gee et. al 2017 – The paradox of weak ties in 55 countries
Park et. al 2018 - The strength of long-range ties in population-scale social networks

2/27: Focal Theory of Networks, Communities/Equivalence/Cohesion

Problem Set 3 Due

Problem Set 4 Assigned

Readings:

Fortunato 2010 (Parts 1-6)
(*More to be determined*)

Borgatti et. al Chapter 11 (Subgroups)
Borgatti et. al Chapter 12 (Structural Equivalence)

Supplemental Readings (Optional):

Feld 1981 – The Focused Organization of Social Ties
Borgatti and Grosser 2015 (Structural Equivalence – Meaning and Measures)
Optional: Borgatti et. al Chapter 6 (Only Read Part on Hierarchical Clustering)
Optional: Fredkin 1984 Structural Equivalence and Cohesion
Optional: Fortunato 2010 (Parts 1-6)

Articles for Presentation (on 3/13):

Cherepnalkoski and Mozeti 2016 – Retweet networks of the European Parliament: evaluation of the community structure
Weng et. al 2010 - To study the technological network by structural equivalence

(3/6: No Class – Spring Break)

3/13: Bipartite Networks and Projections, QAP Models

Final Project Introduced

Readings:

Borgatti et. al Chapter 8 (through 8.5)
Borgatti et. al Chapter 13

(More to be determined)

Articles for Presentation (on 3/20):

To be determined

3/20: Networks, Games, Cascades, Thresholds

Readings:

Chapter 6 - Easley and Kleinberg (NCM)

Chapter 12 - Easley and Kleinberg (NCM)

(More to be determined)

Articles for Presentation (on 3/27):

To be determined

3/27: Models of Contagion, Diffusion, and Influence

Problem Set 4 Due

Problem Set 5 Assigned

Readings:

Chapter 19 - Easley and Kleinberg (NCM)

Chapter 21 - Easley and Kleinberg (NCM)

(More to be determined)

Articles for Presentation (on 4/3):

To be determined

4/3: Stochastic Actor Oriented Models, Models of Network Polarization

Project Proposal Due

Articles for Presentation (on 4/10):

To be determined

4/10: Data Visualization; In-class Workshop on Final Presentations

(Initial) Project Data Due

4/24: In-class presentations

Problem Set 5 Due

Final Project Presentations (Due)

5/11 (Thursday): Final Papers/Projects due to Canvas by 11:59pm